Flight-Validated Digital Filtering for Vibration Mitigation on a 250 kg eVTOL: An FFT-Guided Design Approach

Jonathan Spraggett*, Nima Pourmostaghim†, Younes Sadat-Nejad ‡
MOSTAVIO INC, Toronto, ON, M9L 1V5, Canada

Rotor-induced vibrations corrupt Inertial Measurement Unit (IMU) readings and can destabilize multirotor flight, a risk that grows with vehicle mass. This paper presents a flight-validated, software-only vibration-mitigation strategy for a ~250 kg coaxial quadcopter electric vertical take-off and landing (eVTOL) aircraft. An FFT-guided workflow is used to identify the dominant blade-passage frequency and harmonics, then tune a second-order 30 Hz Butterworth low-pass filter cascaded with two narrow IIR notch filters at 46 Hz and 92 Hz in the gyro feedback path. Indoor hover tests compared mechanically damped, unfiltered, and digitally filtered IMU configurations. The final filter set reduces angular-rate harmonic power by 33.6 ± 0.97 dB (~99.96%), decreases total harmonic RMS angular-rate by 98%, and lowers throttle standard deviation by 60 ± 10%, enabling repeatable 11-minute stable hovers that were not achievable with unfiltered or mechanically isolated configurations. Stability-margin and latency analysis of an approximate roll-rate loop show that the filter chain preserves gain (16.2 dB) and phase (93.9°) margins within standard rotorcraft criteria, introducing only ~16 ms of additional delay, which is less than one quarter of the available time-delay margin. These results highlight the practical impact of digital filtering as a weight-neutral, retrofit-friendly approach that can significantly improve large eVTOL stability and performance.

I. Nomenclature

AAM	=	advanced air mobility
BPF	=	blade passage frequency
CG	=	center of gravity
$EASA$	=	European Union Aviation Safety Agency
$eVTOL$	=	electric Vertical Take-Off and Landing
FAA	=	Federal Aviation Administration
FFT	=	Fast Fourier Transform
IMU	=	Inertial Measurement Unit
IIR	=	Infinite Impulse Response
LPF	=	low-pass filter
PID	=	proportional integral derivative
PSD	=	power spectral density
RMS	=	root-mean-square
RPM	=	revolutions per minute
$STOM$	=	system take-off mass
UAV	=	unmanned aerial vehicle
Δ	=	change / difference
Σ	=	summation operator
σ	=	standard deviation
ω	=	angular rate
$L(s)$	=	open-loop transfer function for the roll-rate loop

*Robotics Software Engineer, MOSTAVIO
†Chief Executive Officer, MOSTAVIO
‡Chief Technology Officer, MOSTAVIO

II. Introduction

Rotor-induced vibrations are a key challenge for reliability, ride quality, and certification in electric vertical take-off and landing (eVTOL) aircraft, particularly as designs scale from small unmanned aerial vehicle (UAV) to passenger-class vehicle [1, 2]. During flight, rotating propellers generate periodic disturbances that appear in the vibration spectrum as peaks at the rotor's blade passage frequency (BPF) and its harmonics [3]. These cyclic loads excite structural resonance and can increase fatigue, noise levels, and the risk of structural wear and failure [4].

Multirotor platforms such as eVTOLs rely on accurate Inertial Measurement Unit (IMU) estimation to maintain flight stability [5]. IMU noise caused by vibration can be mistaken for genuine movement, triggering false control actions that, in extreme cases, can lead to loss of control [4]. At the aircraft level, this manifests as cyclic throttle commands, degraded hover stability, and increased motor loading.

Traditional mitigation strategies for heavy platforms focus on attenuating the vibration before it enters the control loop. These strategies rely on hardware additions (isolation mounts [6], structural redesign [2], rotor phasing [7, 8]) that raise weight, cost, or complexity. While digital filters are routinely employed to reduce vibration-induced IMU noise within the control loop, most published work focuses on low-pass filter (LPF) and Infinite Impulse Response (IIR) notch filtering in small multirotor UAVs (typically < 15 kg) [9, 10] and in simulation or laboratory studies of eVTOL drones [11]. As a result, a gap remains in empirical validation for heavier platforms, particularly in the 100–300 kg eVTOL class studied here.

This paper presents a software-only vibration-mitigation approach that is validated through indoor tethered hover tests on a ~250 kg coaxial quadrotor eVTOL. Our approach uses a data-driven workflow: Fast Fourier Transform (FFT) analysis of IMU data to identify the dominant BPF and harmonics. Followed by the design and implementation of a filter architecture to maximize attenuation of IMU vibration noise while preserving control latency and stability margins, key requirements for certification.

III. Related Work

Vibration mitigation techniques typically fall into three broad categories: digital filtering, passive structural modifications, and active vibration control.

A. Digital Filtering Techniques

Digital filtering methods such as LPF and notch filters are widely used in multirotor UAV autopilots to attenuate rotor-induced vibration noise before it enters the control loops [12]. In ArduPilot, the IMU processing pipeline typically consists of an LPF followed by one or more harmonic notch filters tuned to the rotor BPF [13]. PX4 implements a closely related filter design via its IMU cutoff and notch-filter parameters [14]. The present work adopts a similar structure by combining an LPF with fixed-frequency notch filters in the gyro feedback path.

Low-pass filtering attenuates noise above the cutoff frequency so that high-frequency content, such as vibration-induced noise, does not confuse the controller [15]. However, there is a trade-off; lowering the cutoff frequency increases group delay and phase lag, which can degrade disturbance rejection and stability margins in flight control systems [15, 16]. In one ~10 kg hydrogen tail-sitter UAV (NederDrone), the designers set the attitude-rate LPF cutoff frequency relatively low (1.5 Hz on roll/pitch, 0.5 Hz on yaw) to avoid a structural resonance, at the cost of reduced disturbance rejection performance [9]. This example illustrates the performance penalty of relying solely on aggressive LPF when structural vibrations are severe.

To selectively target vibration-induced noise, notch filters can attenuate a narrow band of frequencies corresponding to the dominant vibration tone, with minimal impact on other frequencies [15]. Chen, L et al. validated this approach by embedding a notch filter in a nonlinear controller on a ~1.1 kg quadcopter, which demonstrated a reduction in vibration-induced noise at the desired frequency and improved attitude stability [10]. A recent 4.8 kg eVTOL project likewise reported the need to account for structural vibration issues during flight tuning. They implemented a notch filter to attenuate structural modes that interfered with the controller [11]. These studies support the use of targeted notch filtering within attitude controllers to suppress specific vibration tones.

Despite these developments, empirical studies validating digital filtering strategies for large-scale eVTOL aircraft remain scarce. NASA's handling qualities guidance for eVTOLs reports that many flight controllers use notch filters to avoid structural mode frequencies [17], indicating that such techniques are known within the large-UAV and eVTOL domains. However, flight-validated case studies are not widely available in the literature. Much of the available knowledge is based on simulations, small-UAV experiments, or anecdotal tuning experience rather than systematic flight tests at higher mass scales.

B. Passive Structural and Isolation Methods

Beyond software methods, a large body of work focuses on mitigating vibrations at the source or along the structural path without active control. Okayama, S., reported a case where the hover revolutions per minute (RPM) of a large quadcopter coincided with the arm torsion mode, causing resonance and severe oscillations [18]. This case highlights a key design guideline: place dominant structural modes outside the operational RPM range. Chen, K. et al. investigated the structural vibrations of a multirotor passenger eVTOL and showed that the dominant bending and torsional modes of the arms were below ~200 Hz, and that adjusting arm geometry could significantly reduce vibration amplitudes in hover [2]. These studies show that structural layout and stiffness directly influence rotor-induced vibration levels.

Another common passive approach is to use vibration isolation mounts to decouple sensitive components from the fuselage and suppress vibration transfer. Isolation systems typically use low-stiffness mounts with damping to lower their natural frequency below the BPF and avoid resonance in the operating envelope [6]. An additional passive technique is the use of tuned mass absorbers or inertial dampers attached to the structure and tuned to oscillate out of phase with the problematic mode. This shows a reduction of structural vibration with modest mass and integration penalties in aerospace applications [19].

While these passive methods can be highly effective at mitigating vibration, they require considerable design and tuning effort to avoid conflicts with the aircraft's structural resonances. Their effectiveness may also be limited if weight or design constraints prevent sufficient damping.

C. Active Vibration Control

Active vibration control strategies seek to counteract vibrations by modifying actuator commands rather than altering structures or sensor processing. Techniques include rotor phasing, in which the relative phase angles of multiple rotors are optimized to minimize periodic loads at the fuselage [7]. Similarly, Cho, C. proposed active motor torque damping to counteract vibration forces and reduce structural vibrations in a simulated tiltrotor eVTOL [8]. Higher harmonic control, historically used in helicopter main-rotor systems, is conceptually related and aims to reduce hub loads by adding harmonics to the blade pitch commands [20]. However, active vibration control methods are inherently actuator-dependent, require additional sensing and control complexity, and currently lack widely documented flight-test validations on large eVTOL platforms. As a result, they complement but do not replace more traditional filtering and structural approaches in current practice.

D. Industry and Certification Context

For passenger-carrying eVTOLs, vibration is not only a control and sensor-fidelity concern but also a certification-level consideration. The European Union Aviation Safety Agency (EASA) Special Condition for small-category VTOL aircraft (SC-VTOL-01) requires VTOL.2160 (Vibration), which states that "each part of the aircraft must be free from excessive vibration throughout the limit flight envelope". This requirement is analogous to rotorcraft vibration requirements and is motivated by structural integrity, fatigue, and passenger comfort considerations [21, 22].

On the Federal Aviation Administration (FAA) side, eVTOL programs such as Joby's JAS4-1 and Archer's M001 are being certified under special class airworthiness criteria that supplement Part 23 with additional structural and noise considerations for eVTOLs [23, 24]. NASA eVTOL white papers similarly emphasize that vibration and acoustic impact are central design constraints for advanced air mobility (AAM) aircraft [25].

These regulatory and industrial developments indicate that managing rotor-induced vibration is a prerequisite for both passenger comfort and certification in AAM. While the ~250 kg platform studied in this work is unmanned, we treat IMU-level mitigation of rotor-induced vibration as a necessary but not sufficient step toward meeting these expectations. Comprehensive structural and cabin-level comfort assessments remain important future work.

E. Identified Gap and Novel Contribution

A literature gap exists regarding the empirical validation of purely software-based vibration mitigation strategies on large-scale eVTOL platforms. Previous published work has primarily focused on structural solutions, small UAV implementations, or simulation-based analyses. Large-scale eVTOL efforts such as Joby and Archer provide minimal public communication on their IMU signal-filtering strategies. To our knowledge, there are no widely available flight-test case studies that quantify software-only IMU filtering for vibration mitigation on multirotor eVTOLs in the 100-300 kg class.

This paper directly addresses that gap by presenting a flight-test-based case study on a ~250 kg heavy-lift eVTOL,

utilizing a filtering approach that builds on practices used in the open-source autopilot community. In addition to describing the filter design, we quantify frequency-domain attenuation, filter-induced latency, and closed-loop stability margins to assess the filtering chain's impact on control performance. The results provide empirical evidence that relatively simple digital filters commonly used on smaller UAVs can scale effectively to large platforms without compromising stability. This offers practical advantages such as reduced complexity, lower weight penalty, and meaningful reductions in rotor-induced vibrations at the IMU.

By providing a detailed, flight-validated analysis of software-based vibration mitigation, this work aims to offer practical guidance for designers and developers operating at similar mass and rotor scales, and to complement ongoing structural and active-control approaches in the broader eVTOL community.

IV. Methodology

A. Aircraft System Description

The experiments were conducted on a coaxial quadcopter eVTOL prototype with a system Take-off mass (STOM) of ~250 kg, as seen in Fig. 1, developed by MOSTAVIO, a Toronto-based aerospace startup specializing in autonomous heavy-lift eVTOLs. MOSTAVIO's platform development process has received external validation as a Stage 1 winner in GoAERO, a NASA and industry-backed global competition for autonomy-enabled emergency-response eVTOL aircraft, and through collaborative applied research with Centennial College on eVTOL structures and battery safety [26–28]. We therefore view the test aircraft as representative of one emerging class of ~250 kg heavy-lift eVTOL platforms, and the IMU filtering methodology presented here is readily applicable to other designers working at similar mass and rotor scales. The platform has a symmetrical layout with four arms, each featuring coaxially arranged motors powering

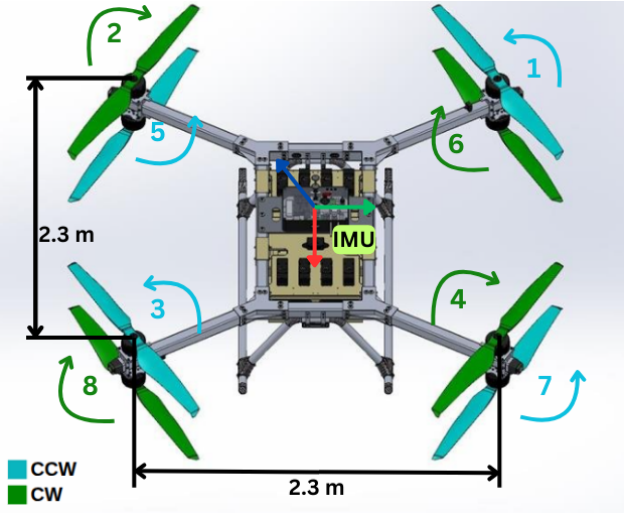


Fig. 1 Coaxial quadrotor eVTOL with rotor direction and IMU Labeled

two-blade propellers measuring 1.57 m in diameter, as summarized in Table 2. This coaxial arrangement alternates the rotation directions of the upper and lower motors to ensure balanced torque production and effective yaw control [29]. Adjacent arms rotate in opposite directions (clockwise and counterclockwise), enhancing stability under normal and failure conditions. The vehicle's symmetry ensures the center of gravity (CG) aligns closely with the geometric center, simplifying control dynamics [28].

B. Data Acquisition and Synchronization

IMU data was sampled at 8 kHz and down-sampled to 800 Hz for logging, which remains safely above twice the highest BPF harmonic analyzed in this study. Data logging utilized a modular container for Arbitrary messages (MCAP) format, with sensor calibration performed before each flight, including gyro recalibration and horizon leveling.

Table 2 Specification of eVTOL

Specification	Value
System Take-off Mass (STOM)	~250 kg
Dimensions (excluding propellers)	2.3 m × 2.3 m × 0.8 m
Propeller Diameter	1.57 m
Frame material	6061-T6 Aluminum
Frame Thickness	Square Tube 3"x3" thickness 1/8"
Maximum Thrust per Rotor	101.45 kgf
Maximum Torque per Rotor	76.97 Nm
Maximum RPM	2203 RPM
IMU Sensor Suite	ICM-20649, ICM-42688P, ICM-20602

C. Flight Test Campaign

We conducted a series of indoor hover tests (7 reported in this paper) with the global positioning system (GPS) and magnetometer disabled, ensuring consistent environmental conditions. The eVTOL movement was limited by being physically tethered to anchors bolted to the floor. The pilot avoided pushing against the restraints, as doing so can affect the IMU data. Other safety measures included protective barriers, ear protection, and radio communication. Flight tests involved performing hover maneuvers at approximately 0.3 m above the ground.

D. Mechanical Damper

We preliminarily evaluated mechanical isolation methods by mounting the IMU on three different configurations. One commonly used for small UAVs, and two 3D-printed mounts using the same open-source CAD design, but different rubber inserts as seen in Fig. 2

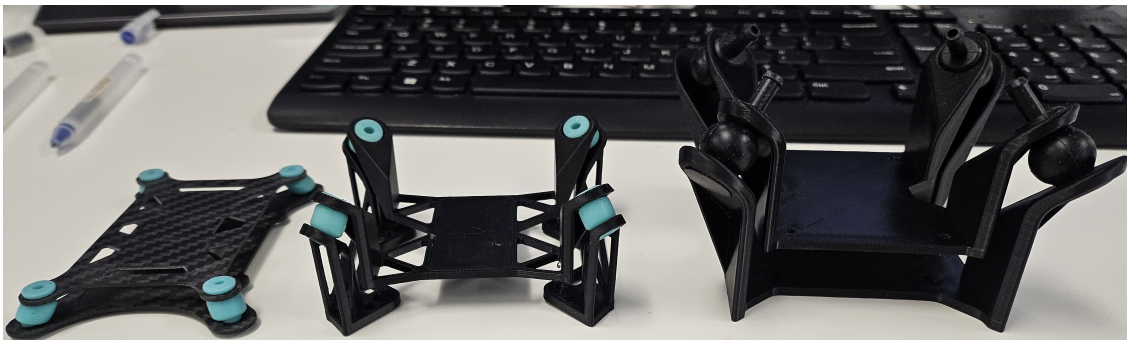


Fig. 2 Physical dampers evaluated. In order: store-bought, 3D-printed blue, 3D-printed black.

E. FFT-Based Filter Design Workflow

Filter parameters were initially chosen from ground tests and iteratively tuned through flight tests based on spectral characterization, targeting vibration peaks observed in the FFT and power spectral density (PSD) analyses shown in Fig. 4. Aliasing effects were minimized by ensuring the sample rate was at least twice the analysis bandwidth [30].

F. Plant Model Derivation

To demonstrate the practicality of this approach, it is essential to show that these filters do not degrade stability to unacceptable levels. To quantify the impact, we analyze the inner attitude-rate loop, implemented at 1 kHz, which is the most critical part of the controller. We present the roll-rate loop for clarity. The pitch-rate loop is symmetric, while the yaw-rate and outer attitude/position loops operate at much lower bandwidth and therefore have larger margins.

We construct an approximate open-loop model of the roll-rate axis by combining the gyro signal chain, the rate controller, and a simplified plant that relates commanded torque to roll rate, consistent with typical motor/airframe dynamics for multirotor quadcopters [31]. This model can be used to generate Bode plots with and without the added filters and calculate stability margins.

To derive the model, we start from Newton–Euler rigid-body dynamics, linearized around hover. Neglecting cross-axis coupling terms, the roll rate dynamics reduce to Eq. (1).

$$I_{xx}\dot{p} = \tau_\phi \tag{1}$$

Where I_{xx} is the roll moment of inertia, $\dot{p}$ is the body roll acceleration and τ_ϕ is the control torque of the body x-axis. For a multirotor, the roll torque is proportional to the difference in rotor thrust [31].

$$\tau_\phi = L(T_r - T_l) \tag{2}$$

Where L is the distance from the rotor to the vehicle CG, and T_r, T_l are the right and left rotor forces, respectively. The thrust of each rotor, $T_i(\sigma)$, can be approximated as a function of the normalized throttle command σ. Linearized around the hover operating point σ_0, we obtain:

$$T_i(\sigma) \approx T_0 + k_T\,(\sigma - \sigma_0), \qquad k_T \triangleq \left.\frac{\partial T_i}{\partial \sigma}\right|_{\sigma_0}. \tag{3}$$

Where $T_0 = T_i(\sigma_0)$ is the thrust around hover and k_T is the thrust slope vs throttle near hover. For a rotor pair with coaxial efficiency $\eta_c \in (0, 1)$, throttle change as $\Delta\sigma = \sigma - \sigma_0$, and four rotors per side, the incremental thrust on the right due to a roll command can be approximated as:

$$\Delta T_r \approx 4\eta_c k_T\,\Delta\sigma \tag{4}$$

Define the input $u \triangleq \Delta\sigma$, and apply $+u$ on the right side and $-u$ on the left side, the control torque becomes

$$\tau_\phi = k_u\,u \qquad k_u = 8\eta_c k_T L \tag{5}$$

Where k_u represents the effective roll torque gain, substituting Eq. (5) into Eq. (1) and taking the Laplace transform yields the linearized roll-axis plant:

$$P_\phi(s) = \frac{P(s)}{U(s)} = \frac{k_u}{I_{xx}s} \tag{6}$$

The roll–rate controller is implemented as a discrete proportional integral derivative (PID). For the following analysis, we approximate this discrete implementation by its continuous-time counterpart $C_\phi(s)$ in Eq. (7):

$$C_\phi(s) = K_P + \frac{K_I}{s} + K_D s \tag{7}$$

The measured roll rate p from the IMU is filtered before being fed to the controller. The signal passes through a second-order Butterworth LPF $F_{\text{LPF}}(s)$, as given by Eq. (8), with f_{cutoff} as the cutoff frequency to start attenuation [32].

$$F_{LPF}(s) = \frac{\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2}, \qquad \omega_c = 2\pi \cdot f_{\text{cutoff}} \tag{8}$$

After the LPF, the signal passes through up to two second-order IIR notch filters $F_N(s)$ using the standard second-order form in Eq. (9) that targets attenuation parameterized by its center frequency f_{center} and bandwidth bw [33]

$$F_{N_c}(s) = \frac{s^2 + \omega_0^2}{s^2 + (\frac{\omega_0}{Q})s + \omega_0^2}, \qquad \omega_0 = 2\pi \cdot f_{\text{center}} \qquad Q = \frac{f_{\text{center}}}{bw} \tag{9}$$

The complete measurement-filtering transfer function is therefore the product of the LPF and both notch filters (Eq. (10)):

$$F(s) = F_{LPF}(s)F_{N_1}(s)F_{N_2}(s) \tag{10}$$

Finally, the open-loop transfer function used for stability-margin evaluation is defined as:

$$\boxed{L(s) = C_\phi(s)F(s)P_\phi(s)} \tag{11}$$

G. Performance Metrics

Primary performance metrics included throttle standard deviation during hover (when throttle is above 50% and has settled), and spectral measures derived from the IMU and throttle PSDs. The PSD was computed via Welch's method [34] with an 800 Hz sample rate, Hann window, 256-sample segments, 50% overlap. Vibration levels were quantified using an integrated PSD band power rather than a single bin peak value to account for RPM fluctuations, windowing effects, and modulation sidebands [35, 36].

For the kth harmonic of the BPF, the band-limited angular-rate power was computed by integrating the PSD over a ± 3 Hz window around $k \cdot f_{\mathrm{BPF}}$:

$$P_k = \int_{k \cdot f_{\mathrm{BPF}} - 3\,\mathrm{Hz}}^{k \cdot f_{\mathrm{BPF}} + 3\,\mathrm{Hz}} \mathrm{PSD}(f)\, df. \tag{12}$$

The total harmonic angular-rate power was then obtained by summing over the first N_h harmonics,

$$P_\Sigma = \sum_{k=1}^{N_h} P_k, \tag{13}$$

with $N_h = 5$ for the IMU data and $N_h = 4$ for the throttle data. The square root of P_Σ yields the total harmonic root-mean-square (RMS) angular-rate metric that can be related to estimator variance and fatigue calculations. Band powers are also reported in decibel units using Eq. (14).

$$P_k^{\mathrm{dB}} = 10 \log_{10}(P_k)\,. \tag{14}$$

V. Results

A. Physical Damper Effectiveness

Initial attempts to fly without software filtering were unsuccessful due to excessive oscillations and throttle saturation. To circumvent noisy IMU data, various physical dampers were evaluated, as summarized in Fig. 3, during an automated ground motor test that raised the input throttle to 40%. All dampers increased angular rate standard deviation ($\omega\sigma$) relative to the rigidly mounted, unfiltered IMU baseline. Notably yaw $\omega\sigma$ increased from 0.0036 rad/s to 0.07 rad/s ($\sim 19x$) in the store bought damper and 0.15 rad/s ($\sim 42x$) in the 3D printed blue damper. These results confirm that, for this airframe, the evaluated dampers amplified vibration rather than attenuating it, consistent with earlier reports of poorly tuned isolators exciting structural modes [18].

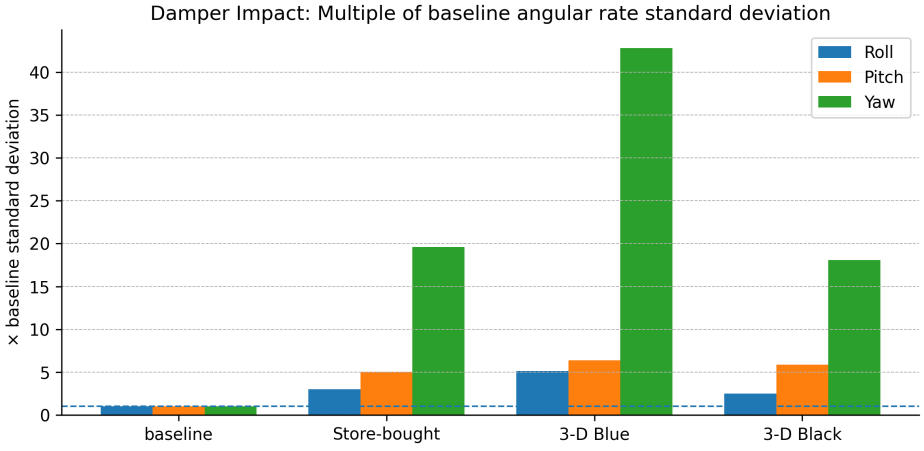


Fig. 3 IMU angular rate standard deviation increase vs damper type on a motor ground test.

B. Vibration Characteristics

Initial flight tests confirmed previous research and identified significant rotor-induced vibrations primarily at the BPF [37], approximately ~46 Hz as seen in Eq. (15).

$$\sim 1380 \text{ RPM} \times \frac{2 \text{ blades}}{60 \text{ s}} = \sim 46 \text{ Hz} \tag{15}$$

An FFT of the raw IMU angular-rate data from the first flight test (F-01), as shown in Fig. 4, shows firm spectral peaks at ~46 Hz and harmonics at integer multiples (~92 Hz, ~138 Hz, etc.). These frequencies fall within the low-frequency band where Chen, K. et al. reported that the dominant arm modes of a large multirotor passenger eVTOL occur below ~200 Hz [2]. Consistent with that observation, we infer that the ~46 Hz peak is likely rotor-induced excitation coupling with one of the airframe's bending or torsional modes. In this work, however, we keep the airframe configuration fixed and focus exclusively on mitigation within the IMU signal chain via digital filtering.

Filtering comprised an LPF and two IIR notch filters applied directly to IMU gyro signals before the PID controller and state estimators.

- LPF: Second-order Butterworth [32] with a cutoff frequency of approximately 30 Hz to preserve aircraft rigid body dynamics while attenuating higher frequency noise.
- Notch Filters: Second-order IIR notch filters [33] centered at the primary BPF (46 Hz) and its first harmonic (92 Hz), selected based on FFT analysis. Bandwidths were tuned narrowly to suppress vibrations without adversely affecting legitimate control dynamics.

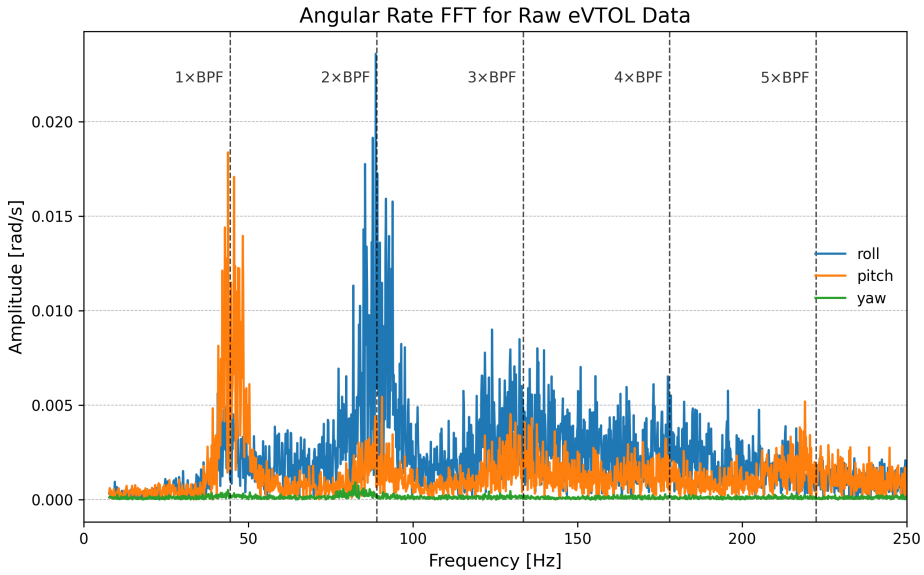


Fig. 4 FFT of raw IMU angular rate during flight F-01 with BPF labeled

C. Effect of Digital Filters

Starting with the first successful hover, from F-01 through F-07 (see Table 3), we examined different filtering elements and their effect on vibration-induced noise reduction, as summarised in Fig. 5, Fig. 6, and Table 4. To verify results,flight F-07 was repeated 7 times ranging from 1 to 11 minute flights.

1. LPF

- F-01 (LPF f_{cutoff} = 30 Hz): Applied a single LPF and resulted in an 11 dB (~92%) reduction, relative to the unfiltered IMU data, in total harmonic angular-rate power (summed BPF band-powers defined in Section IV.G). This also decreased the total harmonic RMS angular-rate from unfiltered 0.15 rad/s to 0.042 rad/s (~72% reduction).

Table 3 Flight campaign filter configuration

Flight ID	LPF (Hz)	Notch #1 (Hz, BW)	Notch #2 (Hz, BW)	Notes
F-01	30	-	-	Baseline, first flight
F-02	20	-	-	Yaw drift
F-03	-	43, 20	-	-
F-04	30	43, 20	-	-
F-05	30	45, 40	-	-
F-06	30	46, 100	92, 100	Roll drift
F-07	30	46, 40	92, 100	Best flight

- F-02 (LPF $f_{\text{cutoff}} = 20$ Hz): We lowered the LPF cutoff frequency, which provided further total harmonic power reduction of 18 dB (~98 %) relative to raw IMU data, but in flight, this configuration exhibited severe yaw instability and required early test termination.

2. Notch Filters

- F-03 (notch $f_{\text{center}} = 43$ Hz, $bw = 20$): used only a notch filter that, when compared with F-01, reduced total harmonic angular-rate power at 46 Hz by 4.8 dB (~67%), but increased power at 92 Hz by 4.9 dB (~207%).

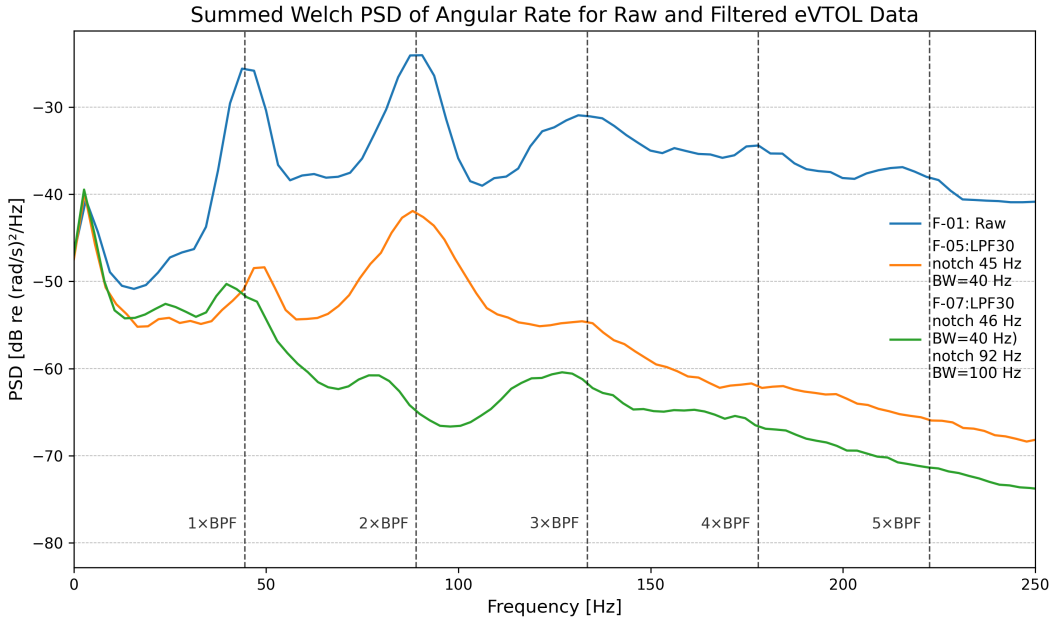


Fig. 5 Welch PSD of IMU gyro data for various filter configurations, illustrating the reduction in vibration peaks with each added filter.

- F-05 (LPF $f_{\text{cutoff}} = 30$ Hz, notch $f_{\text{center}} = 45$ Hz, $bw = 40$): We combined the LPF and notch filter, as well as doubling the bandwidth from 20 Hz in F-04 to 40 Hz. This yielded a 23.1 dB (~99.5%) total harmonic angular-rate power reduction relative to raw IMU data.
- F-06 (LPF $f_{\text{cutoff}} = 30$ Hz, notch 1 $f_{\text{center}} = 46$ Hz, $bw = 100$, notch 2 $f_{\text{center}} = 92$ Hz, $bw = 100$): We increased the bandwidth of the first notch filter and added a second notch filter for the first harmonic of the BPF. This further reduced total angular-rate power by up to 37.7 dB (~99.98%) at 46 Hz and 44.2 dB (~99.996%) at 92 Hz relative to raw IMU data, but introduced noticeable roll drift experienced by the pilot.
- F-07 (LPF $f_{\text{cutoff}} = 30$ Hz, notch 1 $f_{\text{center}} = 46$ Hz, $bw = 40$, notch 2 $f_{\text{center}} = 92$ Hz, $bw = 100$): Stable performance was obtained by reducing the first notch filter bandwidth back to 40 Hz. This resulted in a total

harmonic angular-rate power reduction of 33.6 ± 0.97 dB (~99.96%) over 7 flights, relative to the unfiltered IMU data and 19.6 dB (~98.9%) relative to F-01. The total harmonic RMS angular-rate dropped to 0.0052 ± 0.0012 rad/s from raw data with 0.25 ± 0.07 rad/s (~98% reduction) and F-01 0.042 rad/s (~90% reduction).

Table 4 Important harmonic angular rate power reduction from filter tuning

Flight ID	Power at 46 Hz ($\triangle dB$)	Power at 92 Hz ($\triangle dB$)	Total Harmonic Power ($\triangle dB$)	Total harmonic RMS (rad/s)
F-01	-7.1	-19.8	-10.9	0.042
F-05	-29.5	-27.2	-23.1	0.013
F-07	-47.8	-41.7	33.6 ± 0.97	0.0052 ± 0.0012

Main results are summarized by Table 4, where all $\triangle dB$ values are referenced to the unfiltered raw data unless otherwise specified. Overall, the progression from using only a LPF (F-01) through one notch (F-03 to F-05) to the two notch configuration (F-06, F-07) shows a progressive reduction in total harmonic angular rate power as can be seen in Fig. 5 and Fig. 6, with F-07 achieving the most considerable attenuation while retaining acceptable levels of drift.

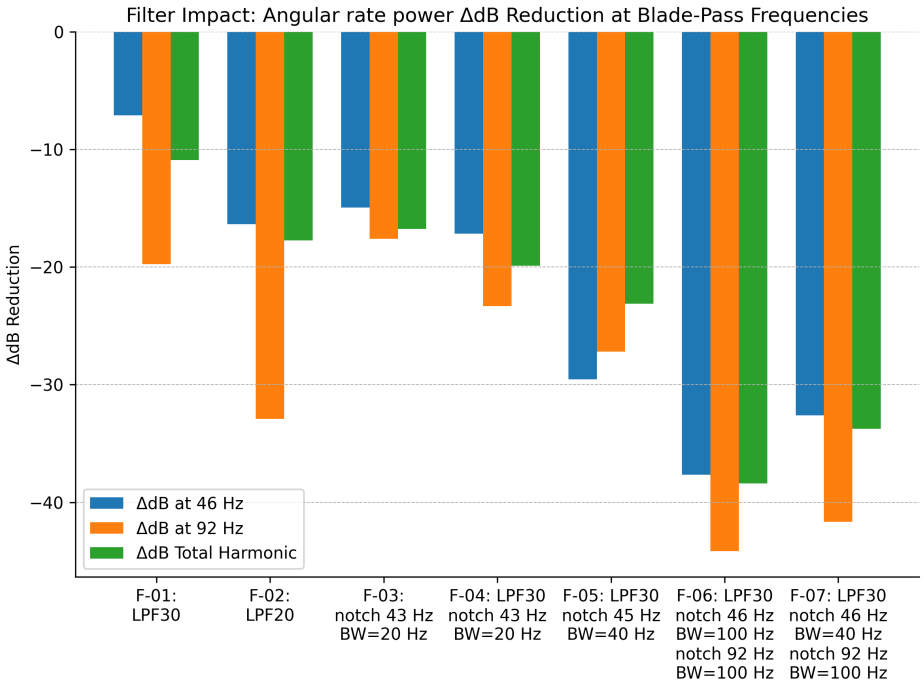


Fig. 6 △dB of angular-rate harmonic power for various filter configurations, illustrating the reduction in vibration peaks with each new flight test.

D. Flight Control and Throttle Stability

Filtering significantly improved flight stability, summarized in Table 5, where the results are relative to flight F-01.

- F-01 (LPF f_{cutoff} = 30 Hz): Reduced mean per-axis angular-rate standard deviation across roll, pitch, and yaw by 48.3% relative to raw IMU data, significantly improving throttle standard deviation.
- F-03 (notch f_{center} = 43 Hz, bw = 20): Increased throttle harmonic power by 3.2 dB (~108%) and throttle standard deviation by 19.8% relative to F-01, adversely affecting control.
- F-07 (LPF f_{cutoff} = 30 Hz, notch 1 f_{center} = 46 Hz, bw = 40, notch 2 f_{center} = 92 Hz, bw = 100): The final filter configuration, achieved the best stability metrics, reduced throttle harmonic power by 17.5 dB (~98.2%), and throttle standard deviation by 60 ± 10% relative to F-01. Flight testing indicated reduced control variance and improved hover stability.

Table 5 Throttle Stability Data

Flight ID	Throttle $\triangle\sigma$ (%)	Total Harmonic Throttle power ($\triangle dB$)
F-01	0.0	0
F-02	-22.4	-10.8
F-03	19.8	3.2
F-04	-13	-2.98
F-05	-52.2	-8.3
F-06	-67.8	-24.9
F-07	-60 ± 10	-17.5

E. Stability Margin Verification

Digital filters can introduce additional phase lag near the gain crossover frequency, affecting closed-loop stability [38]. The open-loop transfer function, Eq. (11), from Section IV.F, can be calculated with the selected filters from flight F-07 and values from Table 6 which were derived from the CAD model and static thrust bench tests at hover operating points. The transfer function was then used to create the Bode plot shown in Fig. 7. This allowed us to compute the gain and phase margins at the crossover frequency to quantify the incremental phase lag introduced by the filters. This model is sufficient because the objective is to verify that the proposed filtering does not significantly reduce stability margins relative to the baseline controller.

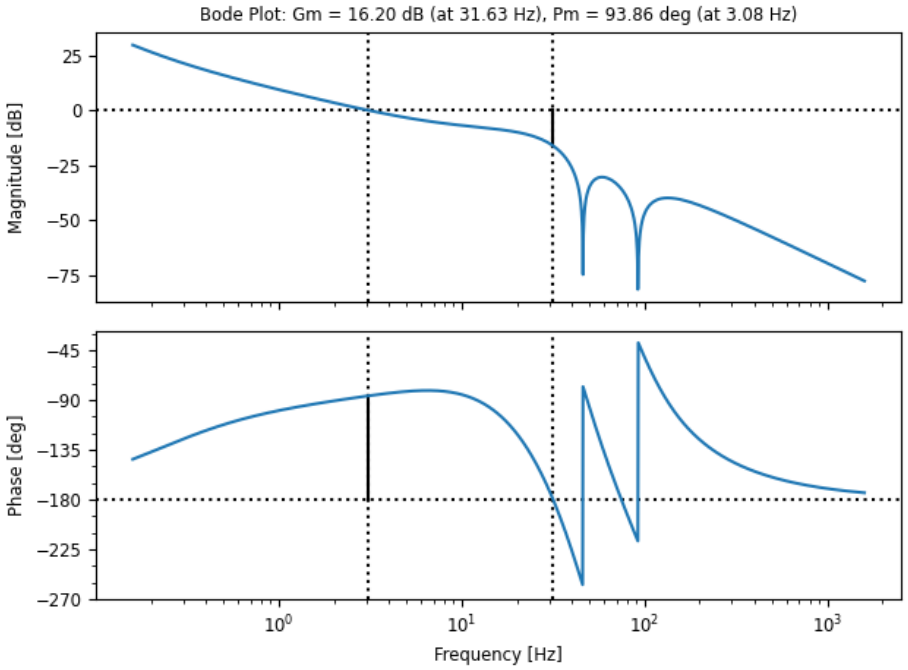


Fig. 7 Bode plot of the derived plant model with filters from F-07. Gain (16.2 dB) and phase (93.9°) margins drawn in black.

Prior to adding the filters, the roll-rate loop exhibited an infinite gain margin and a phase margin of 107.7°. With the proposed filtering, the gain margin is reduced to 16.2 dB and the phase margin to 93.9°. These values remain comfortably above the rotorcraft criteria for military aircraft as specified in SAE Aerospace Standard AS94900A [39] with a gain margin ≥ 6 dB, phase margin ≥ 45°. These margins reduce the risk of adverse coupling with structural modes, and a full handling-qualities assessment is left for future work.

The incremental phase lag introduced by the digital filters is small enough that the controller retains adequate

Table 6 Plant Components

Component	Value
I_{xx}	$124.406[kg \cdot m^2]$
L	$1.626[m]$
η_c	0.82
Thrust slope k_T	$1435.792[N/\text{unit throttle}]$
K_P	0.15
K_I	0.2
K_D	0.003

robustness while achieving substantial vibration reduction, supporting compliance with the intent of EASA SC-VTOL.2160 [21, 22]. A more detailed loop-shaping study using a fully identified airframe and actuator model is left for future work.

F. Latency Verification

To quantify the latency introduced by the proposed filters, we discretized $F(s)$ Eq. (10) at the onboard sampling rate (1 kHz) for the attitude rate controller. The group delay was evaluated numerically using frequency-response analysis as seen in Fig. 8. The filter chain introduces a group delay of at most $T_D \approx 16.45$ ms. From a handling-qualities

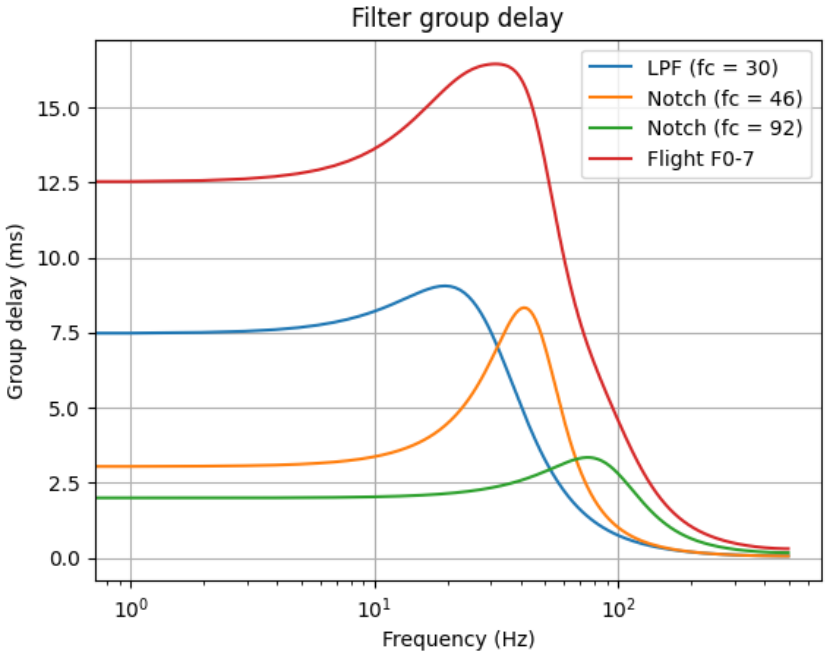


Fig. 8 Group delay plot showing the filter's group delay.

perspective, the relevant metric is not the absolute group delay but its impact on closed-loop stability. Using the gain and phase margins from Section V.E, this corresponds to a time-delay margin of $T_{\text{DM}} = \phi_m/\omega_{\text{gc}} = 84.7$ ms with ϕ_m as the phase margin and ω_{gc} as the gain crossover frequency both in radians [40]. The filters consume less than a quarter of the available time-delay margin at the roll-rate crossover frequency, leaving substantial robustness to additional latencies (e.g., computation, communication, actuator dynamics).

We do not claim compliance with specific handling-qualities levels (e.g., ADS-33E bandwidth/phase-delay criteria), but note that retaining classical stability margins and bounding the added latency is consistent with the underlying

stability and latency considerations in ADS-33E bandwidth/phase-delay criteria and recent multirotor handling-qualities studies [41, 42]. In this sense, the proposed filtering is trading a small fraction of the available time-delay budget for a significant reduction in rotor-induced vibration at the IMU.

VI. Discussion

A. Software vs. Hardware Vibration Attenuation Performance

Flight tests demonstrated that significant vibration-induced noise reduction is achievable solely through digital filtering, even in heavy (250 kg-class) eVTOLs. Using a combination of tuned notch filters at the BPF and a complementary LPF, the gyro signal chain is shaped so that the dominant vibration-induced noise is rejected from the feedback path while preserving the frequencies required for attitude control. In contrast, all tested physical dampers increased the yaw rate standard deviation; this behavior is consistent with resonance issues reported on large quadcopters when structural modes coincide with rotor excitation frequencies [18]

The software-based approach also avoids the weight and structural complexity of passive methods, as well as the excessive control latency associated with overly aggressive low-pass filtering [15]. Even after achieving flight in F-01, the eVTOL still exhibited noticeable throttle oscillations and a higher throttle standard deviation relative to flight F-07, which degraded flight stability. To further reduce vibration power, we set the gyro LPF cutoff frequency to 20 Hz during flight F-02, resulting in noticeable yaw drift. In flight F-03, we introduced a notch filter at the rotor BPF to prevent that narrow-band noise from entering the control feedback path. However, the initial tuning increased throttle harmonic power and variance, motivating the subsequent bandwidth adjustments in F-05 and F-07.

Consequently, the combined filtering approach delivered a practical compromise between vibration-induced noise attenuation and closed-loop robustness. The filters were tuned under explicit stability-margin and latency constraints so that vibration at the BPF was suppressed without eroding the controller's gain/phase margins or consuming more than a modest fraction of the available time-delay budget. This filter configuration targeted BPF, while preserving most of the control bandwidth that an overly aggressive LPF, as seen in F-02, would have diminished. This explains why the digital filters outperformed the physical dampers in reducing IMU vibration noise and improving overall flight performance, even though they do not physically remove airframe vibration.

B. Implications for Control Stability and Sensor Health

Purely digital filtering does not physically eliminate airframe vibration; it substantially reduces the propagation of IMU vibration-induced noise into the control and state estimation loops. This enhances flight stability as seen by a $60 \pm 10\%$ decrease in throttle standard deviation between the first successful flight (F-01) and the final filtered configuration (F-07). Since throttle standard deviation is directly tied to cyclic actuator effort, this reduction is consistent with smoother, more stable hover performance and reduced cyclic loading on the propulsion and avionics structures. Pilots reported steadier hovers, and logged control inputs showed reduced corrective action to compensate for vibration-induced drift.

Though our study did not directly measure component wear, it is known that high-frequency vibrations can cause issues such as connector fretting [43], calibration drift [44], or component fatigue on circuit boards negatively affecting sensor longevity [45]. By reducing high-frequency throttle oscillations, the proposed filters limit the amount of vibration entering the system through rapid actuator commands. In summary, we interpret the observed decrease in throttle variability as an indicator of lower dynamic loading on avionics hardware, which may improve the long-term integrity of the sensing and control system and is important for safe operation.

The stability-margin and latency analysis clarifies why this reduction in throttle variability did not come at the expense of fragile or sluggish control. The roll-rate loop model showed that, even with the entire filter chain in place, the system maintained gain and phase margins comfortably above typical rotorcraft criteria, along with a large time-delay margin. The design is reallocating a small portion of inherent robustness to reject narrowband rotor-induced disturbances at the IMU. This provides additional confidence that the observed improvements in hover stability and actuator loading are sustainable as the platform and control software evolve.

A key limitation of the current implementation is that the notch filters are fixed in frequency and tuned to a BPF derived from static RPM estimates. To accommodate moderate RPM variations during hover and low-speed flight, we employ relatively wide notch bandwidths rather than adapting the center frequency online. This is adequate for the operating points tested here. However, under changing flight conditions (climb, gust response, varying payload), the

RPM can deviate enough to reduce attenuation at the true BPF. Similar frequency-mismatch issues motivating the use of adaptive notch filters have been reported in other high-speed mechanical systems [46]. In future work, we plan to integrate RPM adaptive notch filtering driven by ESC telemetry, similar to existing open-source autopilot practices, to improve filtering robustness [13].

C. Structural Fatigue Implications

Although we did not directly measure structural strain or modal responses, the observed change in throttle variation between the first flight (F-01) and the final filtered configuration (F-07) has implications for structural fatigue. The reduction in throttle standard deviation implies a smaller amplitude of cyclic loading transmitted to the airframe, mounts, and avionics enclosures. The remaining structural vibrations remain, but the excitation imparted by the control system is less energetic due to the smoother throttle behavior.

Classical Basquin-type S–N behavior for aluminum alloys qualitatively indicates that even moderate decreases in stress amplitude can yield substantial increases in fatigue life [47, 48]. However, a rigorous life prediction for this airframe would require detailed structural models, measured strain histories, and mission-level load spectra. In this work, we therefore interpret the observed reductions in throttle variability and high-frequency control effort as providing a conservative margin against fatigue-related issues in avionics structures and connections, and as a pragmatic complement to future structural modifications and component-level fatigue testing.

D. Comparisons to Prior Work and Generalizability

Software alone cannot eliminate all physical vibration; it can only limit how much of it is sensed and acted on by the control system. A combined strategy is ideal, using digital filters to prevent control-induced excitations with targeted structural measures (e.g., IMU-mount stiffening/damping, component relocation, and arm/frame redesign informed by FEA and experimental modal analysis similar to Chen, K. et al. (2023) [2]), which we leave to future work. Our results show that even without structural redesign, shaping the feedback path can smooth throttle commands and reduce cyclic actuator loading, both of which are directly relevant to component longevity.

These results complement rotor-level hardware/control solutions such as relative rotor phasing, which suppressed composite vibration indices by 62% in multirotors [7]. Unlike phasing or redesigned arms, digital filtering is retrofit-friendly, weight-neutral, and configuration-agnostic, making it immediately deployable on existing fleets.

Demonstrating this scalability on a heavy platform provides an initial bridge between anecdotal success in small-UAV communities (e.g., ArduPilot) and the demands of full-scale eVTOLs, positioning digital filtering as a core component of holistic vibration management. The methodology we used, which identifies dominant frequencies, is not specific to this aircraft. Any multirotor with a clear BPF could benefit from this approach. Of course, tuning the cutoff and notch frequencies to the specific aircraft's dynamics is required. However, the FFT-guided workflow can be applied to any new platform to find the desired values systematically. Our approach assumes that the problematic frequencies are consistent (e.g., tied to rotor RPM). If the rotor speed varies significantly in flight, a fixed-notch filter might lose effectiveness; hence, an adaptive filter would be needed, as noted in future work.

In terms of regulatory standards, Radio Technical Commission for Aeronautics (RTCA) DO-160G includes rigorous vibration tests to ensure avionics reliability [49]. International Organization for Standardization (ISO) 2631 defines acceptable levels of whole-body vibration for human comfort and health [50]. Prior work has shown that a 20 dB reduction in vibration could be the difference between a harsh, bumpy ride and a smooth one [51]. While our eVTOL is unmanned, the trend in advanced air mobility is toward larger vehicles that will carry passengers or fragile cargo [52]. In this work, we interpret the targeted attenuation at BPF in the filter design, together with the observed smoothing of throttle commands, as a reduction in narrowband rotor-induced loading at the avionics level, a necessary but not sufficient condition for improved passenger comfort. Quantitative mapping from these avionics-level metrics to ISO 2631 ride-comfort measures remains open, and we identify formal ride-comfort quantification as important future work. By explicitly verifying that the filters preserve gain, phase, and time-delay margins, the proposed design also aligns with the control-system robustness and latency considerations embedded in rotorcraft-style handling-qualities and eVTOL certification frameworks, even though a full handling-qualities assessment is left to future work. This emphasizes weight, space, and flight stability. Digital filtering is therefore a promising approach to managing vibration exposure for both avionics equipment and potential human occupants, facilitating easier compliance and potentially helping meet certification requirements related to structural integrity and occupant comfort. However, full compliance still requires dedicated environmental and ride-quality testing.

VII. Conclusion

To our knowledge, this is among the first publicly reported, flight-validated demonstration that software-only vibration mitigation can stabilize a ~250 kg-class eVTOL. A 30 Hz second-order Butterworth LPF combined with two narrow IIR notch filters at the BPF (46 Hz) and its first harmonic (92 Hz), reduced angular-rate power at the dominant harmonics by 33.6 ± 0.97 dB (~99.96%), lowered total harmonic RMS angular rate by 98%, and decreased throttle standard deviation by 60 ± 10%, enabling repeatable 11-minute stable hovers that were not attainable with unfiltered or mechanically isolated configurations.

Stability-margin and latency analyses show that these benefits are achieved without compromising closed-loop robustness. The filtered roll-rate loop retains a 16.2 dB gain margin and a 93.9° phase margin, comfortably above classical rotorcraft criteria, and the resulting 16.45 ms group delay consumes less than one quarter of the available time-delay margin at the crossover frequency. In effect, the filter chain trades a small portion of the allowable delay budget for a significant reduction in rotor-induced vibration at the IMU and smoother control effort.

Together, the flight data and loop-analysis results establish that digital filtering techniques proven on small UAVs can scale effectively to mid-scale eVTOLs as a retrofit-friendly, weight-neutral complement to mechanical dampers and more complex active vibration controls. Beyond improving control fidelity and sensor longevity, the associated reduction in cyclic actuator loading is consistent with increased structural fatigue margins and reduced avionics vibration exposure, supporting the broader goals of reliability, maintainability, and ride quality in AAM aircraft.

Future work will extend this methodology to RPM-adaptive notch filters driven by ESC telemetry, joint tuning of filters and PID gains, and validation across more aggressive maneuvers, operating points, and airframe geometries to further generalize the approach for the broader eVTOL community.

In summary, the study demonstrates a lightweight, purely digital vibration mitigation strategy that addresses a critical vibration issue in large eVTOLs while preserving conventional stability margins, providing a practical contribution toward safer, more reliable AAM vehicles.

Acknowledgments

The authors would like to express their sincere gratitude to all members of MOSTAVIO, whose collective efforts and contributions were essential to completing this work. Special thanks are extended to Hossein Kazemi for his invaluable assistance in generating the CAD models and maintaining the eVTOL, and to Mostafa Asadi Khanouki for his expertise and support in co-piloting the eVTOL.

References

[1] Silva, C., Johnson, W. R., Solis, E., Patterson, M. D., and Antcliff, K. R., "VTOL Urban Air Mobility Concept vehicles for technology development," *2018 Aviation Technology, Integration, and Operations Conference*, 2018. https://doi.org/10.2514/6.2018-3847.

[2] Chen, K., Meng, W., Wang, J., Liu, K., and Lu, Z., "An investigation on the structural vibrations of multi-rotor passenger drones," *International Journal of Micro Air Vehicles*, Vol. 15, 2023, p. 17568293231199097. https://doi.org/10.1177/17568293231199097, URL https://doi.org/10.1177/17568293231199097.

[3] Balcerak, J. C., "Parametric study of the noise produced by the interaction of the main rotor wake with the tail rotor," 1976. URL https://api.semanticscholar.org/CorpusID:109765931.

[4] de Jesús Villalobos, J., Rojo, E., Ledezma, D., and Lopez, P., "Structural Analysis Based on Frequency Response of a Quadrotor UAV," *Preprints*, 2024. https://doi.org/10.20944/preprints202405.1686.v1, URL https://doi.org/10.20944/preprints202405.1686.v1.

[5] Lv, M., Fan, B., Fang, J., and Wang, J., "Autonomous Landing of Quadrotor Unmanned Aerial Vehicles Based on Multi-Level Marker and Linear Active Disturbance Reject Control," *Sensors*, Vol. 24, No. 5, 2024. https://doi.org/10.3390/s24051645, URL https://www.mdpi.com/1424-8220/24/5/1645.

[6] Loewen, H., "Isolating Components from UAV Vibration," Tech. rep., MicroPilot Inc., Stony Mountain, Manitoba, Canada, 2013. URL https://www.micropilot.com/pdf/isolating-components-uav-vibration.pdf.

[7] Niemiec, R., Gandhi, F., and Kopyt, N., "Relative rotor phasing for multicopter vibratory load minimisation," *The Aeronautical Journal*, Vol. 126, No. 1298, 2022, p. 710–729. https://doi.org/10.1017/aer.2021.94.

[8] Cho, C., "Active Vibration Control Using Electric Rotor Torques in eVTOL Aircraft," Phd dissertation, The Pennsylvania State University, University Park, PA, USA, Aug. 2024. URL https://etda.libraries.psu.edu/catalog/19721cmc7589.

[9] De Wagter, C., Remes, B., Smeur, E., van Tienen, F., Ruijsink, R., van Hecke, K., and van der Horst, E., "The NederDrone: A hybrid lift, hybrid energy hydrogen UAV," *International Journal of Hydrogen Energy*, Vol. 46, No. 29, 2021, pp. 16003–16018. https://doi.org/10.1016/j.ijhydene.2021.02.053.

[10] Chen, L., Zhang, J., and Quan, Q., "Quadcopter attitude control with vibration reduction by additive-state-decomposition dynamic inversion design with a notch filter," *Nonlinear Dynamics*, Vol. 111, No. 9, 2023, p. 8313–8327. https://doi.org/10.1007/s11071-023-08272-6.

[11] Martinelli, E., "Modelling, Control, Integration and Testing of an eVTOL Drone," Master's thesis, Politecnico di Milano, Milano, Italy, Jul. 2022. URL https://hdl.handle.net/10589/190621.

[12] Kim, H., Bandyopadhyay, R., Ozmen, M. O., Celik, Z. B., Bianchi, A., Kim, Y., and Xu, D., "A Systematic Study of Physical Sensor Attack Hardness," *2024 IEEE Symposium on Security and Privacy (SP)*, 2024, pp. 2328–2347. https://doi.org/10.1109/SP54263.2024.00143.

[13] ArduPilot Development Team, "Managing Gyro Noise with the Dynamic Harmonic Notch Filters," , 2024. URL https://ardupilot.org/copter/docs/common-imu-notch-filtering.html.

[14] PX4 Autopilot Development Team, "MC Filter Tuning & Control Latency," , 2025. URL https://docs.px4.io/main/en/config_mc/filter_tuning.html.

[15] Steffensen, R., Steinert, A., Mbikayi, Z., Raab, S., Angelov, J., and Holzapfel, F., "Filter and sensor delay synchronization in incremental flight control laws," *Aerospace Systems*, Vol. 6, No. 2, 2023, p. 285–304. https://doi.org/10.1007/s42401-022-00186-2.

[16] Alam, M., and Rohac, J., "Adaptive Data Filtering of Inertial Sensors with Variable Bandwidth," *Sensors*, Vol. 15, No. 2, 2015, pp. 3282–3298. https://doi.org/10.3390/s150203282, URL https://www.mdpi.com/1424-8220/15/2/3282.

[17] Klyde, D., Mitchell, D., Shubert, M., and Jones, M., "Handling Qualities Test Guide for Powered Lift VTOL Capable Aircraft with Indirect Flight Controls and Operating in Day Time Visual Flight Rules," , Mar 2024. URL https://rosap.ntl.bts.gov/view/dot/74025, tech Report.

[18] Okayama, S., "Torsional vibration on a large quadcopter," *AIAA SCITECH 2023 Forum*, 2023. https://doi.org/10.2514/6.2023-1703.

[19] Scheidler, J. J., and Asnani, V. M., "A Review of Noise and Vibration Control Technologies for Rotorcraft Transmissions," 2016. URL https://api.semanticscholar.org/CorpusID:13810252.

[20] NGUYEN, K., and CHOPRA, I., "Application of higher harmonic control (HHC) to hingeless Rotor Systems," *30th Structures, Structural Dynamics and Materials Conference*, 1989. https://doi.org/10.2514/6.1989-1215.

[21] European Union Aviation Safety Agency, "Special Condition for Small-Category VTOL Aircraft (SC-VTOL-01)," Tech. rep., EASA, 07 2019. URL https://www.easa.europa.eu/en/document-library/special-conditions/sc-vtol-01.

[22] European Union Aviation Safety Agency, "Means of Compliance with the Special Condition VTOL (MOC SC-VTOL)," Tech. rep., EASA, 05 2021. URL https://www.easa.europa.eu/en/document-library/guidance-materials/moc-sc-vtol.

[23] Federal Aviation Administration, "Airworthiness Criteria for the Joby Aero, Inc. Model JAS4-1 Powered-Lift," , Mar 2024. URL https://www.federalregister.gov/d/2024-04690.

[24] Federal Aviation Administration, "Airworthiness Criteria: Special Class Airworthiness Criteria for the Archer Aviation, Inc. Model M001 Powered-Lift," , may 2024. URL https://www.federalregister.gov/d/2024-11192.

[25] Doo, J., Pavel, M., Didey, A., Hange, C., Diller, N., Tsairides, M., Smith, M., Bennet, E., Bromfield, M., and Mooberry, J., *NASA Electric Vertical Takeoff and Landing (eVTOL) Aircraft Technology for Public Services – A White Paper: NASA Transformative Vertical Flight Working Group 4 (TVF4)*, NASA, 2021.

[26] GoAero, "Meet mostavio: Stage 1 winner developing AeroGuardian for emergency response evtol," , May 2025. URL https://www.goaeroprize.com/post/meet-the-teams-mostavio-stage1-winner.

[27] College, C., "Testing before flight: Centennial partners with Mostavio for Evtol Battery Impact Studies," , 2025. URL https://www.centennialcollege.ca/applied-research-innovation-and-entrepreneurship-blog/2025/june/13/testing-before-flight-centennial-partners-with-mostavio-for-evtol-battery-impact-studies/.

[28] Asadi Khanouki, M., Sadat-Nejad, Y., and Pourmostaghimi, N., "Simulation-based evaluation of Motor Failure Resilience in a prototyped multirotor evtol," *Proceedings of the Vertical Flight Society 81st Annual Forum and Technology Display*, 2025. https://doi.org/10.4050/f-0081-2025-218.

[29] Yoon, S., Chan, W. M., and Pulliam, T. H., "Computations of torque-balanced coaxial rotor flows," *55th AIAA Aerospace Sciences Meeting*, 2017. https://doi.org/10.2514/6.2017-0052.

[30] Nyquist, H., "Certain Topics in Telegraph Transmission Theory," *Transactions of the American Institute of Electrical Engineers*, Vol. 47, No. 2, 1928, pp. 617–644. https://doi.org/10.1109/T-AIEE.1928.5055024.

[31] Bresciani, T., "Modelling, Identification and Control of a Quadrotor Helicopter," Msc thesis, Lund University, Department of Automatic Control, Lund, Sweden, 2008. URL http://lup.lub.lu.se/student-papers/record/8847641.

[32] Butterworth, S., "On the Theory of Filter Amplifiers," *Experimental Wireless & the Wireless Engineer*, Vol. 7, 1930, pp. 536–541.

[33] Mahdiani, S., Jeyhani, V., and Vehkaoja, A., "A Review of Transient Suppression Methods of IIR Notch Filters Used for Power-Line Interference Rejection in ECG Measurement," *XIV Mediterranean Conference on Medical and Biological Engineering and Computing 2016*, edited by E. Kyriacou, S. Christofides, and C. S. Pattichis, Springer International Publishing, Cham, 2016, pp. 151–156.

[34] Welch, P., "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," *IEEE Transactions on Audio and Electroacoustics*, Vol. 15, No. 2, 1967, pp. 70–73. https://doi.org/10.1109/TAU.1967.1161901.

[35] Lößle, F., Kostek, A., Schwarz, C., and Schmid, R., "Experimental Measurement of a UAV Rotor's Acoustic Emission," *New Results in Numerical and Experimental Fluid Mechanics XIII*, edited by A. Dillmann, G. Heller, E. Krämer, and C. Wagner, Springer International Publishing, Cham, 2021, pp. 387–396.

[36] Bendat, J. S., and Piersol, A. G., *Random Data: Analysis and Measurement Procedures*, 4th ed., John Wiley Sons, Inc, 2010, p. 1–40.

[37] Pätzold, F., Bauknecht, A., Schlerf, A., Sotomayor Zakharov, D., Bretschneider, L., and Lampert, A., "Flight Experiments and Numerical Simulations for Investigating Multicopter Flow Field and Structure Deformation," *Atmosphere*, Vol. 14, No. 9, 2023. https://doi.org/10.3390/atmos14091336, URL https://www.mdpi.com/2073-4433/14/9/1336.

[38] Singh, G. K., Deodhare, G., Patel, V. V., and Chetty, S., "Automated process for optimal structural notch-filter design," *Journal of Guidance, Control, and Dynamics*, Vol. 31, No. 5, 2008, p. 1504–1508. https://doi.org/10.2514/1.36767.

[39] SAE International Technical Standard, "Aerospace - Flight Control Systems - Design, Installation and Test of Piloted Military Aircraft, General Specification For," SAE Standard AS94900, 2007. URL https://doi.org/10.4271/AS94900.

[40] Horn, J. F., "Non-Linear Dynamic Inversion Control Design for Rotorcraft," *Aerospace*, 2019. URL https://api.semanticscholar.org/CorpusID:88487883.

[41] Baskett, B., "Aeronautical Design Standard Performance Specification Handling Qualities Requirements For Military Rotorcraft," 2000. URL https://api.semanticscholar.org/CorpusID:108220976.

[42] Atci, K., Jones, M. A., and Jusko, T., "Assessment of the Handling Qualities of Multirotor Configurations using Real Time Simulation," 2021. URL https://api.semanticscholar.org/CorpusID:244197192.

[43] Zhang, F., and Flowers, G. T., "Fretting corrosion in electric connectors induced by axial vibration," *2014 IEEE 60th Holm Conference on Electrical Contacts (Holm)*, 2014, pp. 1–8. https://doi.org/10.1109/HOLM.2014.7031055.

[44] Fang, X., Dong, L., Zhao, W.-S., Yan, H., Teh, K. S., and Wang, G., "Vibration-Induced Errors in MEMS Tuning Fork Gyroscopes with Imbalance," *Sensors*, Vol. 18, No. 6, 2018. https://doi.org/10.3390/s18061755, URL https://www.mdpi.com/1424-8220/18/6/1755.

[45] Kim, K.-W., Park, J.-H., Park, T.-Y., and Oh, H.-U., "Experimental Evaluation of the Effectiveness of the Printed Circuit Board Strain-Based Methodology in Space-Borne Electronics with Vertically Mounted Printed Circuit Boards," *Aerospace*, Vol. 11, No. 7, 2024. https://doi.org/10.3390/aerospace11070562, URL https://www.mdpi.com/2226-4310/11/7/562.

[46] de Fretes, R., "Frequency Variable Adaptive Control Applied to Industrial High-Speed Cut Sheet Inkjet Printing Systems," Master's thesis, Eindhoven University of Technology, 2019. URL https://research.tue.nl/en/studentTheses/frequency-variable-adaptive-control-applied-to-industrial-high-sp/.

[47] Basquin, O. H., *The Exponential Law of Endurance Tests*, American Society for Testing and Materials Proceeding, 1910, Vol. 10, pp. 625–630.

[48] Stephens, R. I., Fatemi, A., Stephens, R. R., and Fuchs, H. O., *Metal Fatigue in Engineering*, 2nd ed., John Wiley & Sons, New York, NY, 2000.

[49] Sweeney, D. L., *Digital Avionics Handbook*, 3rd ed., CRC Press, 2015, pp. 11–1–11–22.

[50] International Organization for Standardization, "ISO 2631-1: 1997, Mechanical Vibration and Shock - Evaluation of Human Exposure to Whole-body Vibration," , May 1997.

[51] Chen, Y., Zhang, S., Chang, J., Fereidooni, A., and Wickramasinghe, V., "Development of a multiaxis active seat mount to mitigate helicopter aircrew whole-body vibration exposure," *Journal of Intelligent Material Systems and Structures*, Vol. 30, No. 17, 2019, pp. 2544–2555. https://doi.org/10.1177/1045389X19844027, URL https://doi.org/10.1177/1045389X19844027.

[52] NASA, "Advanced Air Mobility Air Taxis, Drones and You," , Oct. 2024. URL https://www.nasa.gov/wp-content/uploads/2024/10/aam-litho-oct-2024-v2.pdf?emrc=67790cb942a02.